

Short-Term Temperature Forecasting Using LSTM-CatBoost Combination Method

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Abstract—The Internet of Things (IoT) is a fast-growing technology that has the potential to revolutionize various aspects of our daily life, such as weather forecasting. Accurate weather forecasting allows people to prepare for sudden weather changes, reducing property damage caused by weather disasters. Traditional long short-term memory (LSTM) is good at modeling long-term dependencies, but it has the problem of overfitting. In view of this, categorical boosting (CatBoost), a machine learning model having strong forecasting ability and not easy to overfit, is employed to enhance weather forecasting accuracy. By combining these two powerful and heterogeneous models, an LSTM-CatBoost averaging method, which has both low variance error and bias error in principle, is proposed to forecast weather-related variables, e.g., temperature. This study uses the data from the IoT-based automatic weather station deployed across Botswana to conduct experiments, and the results show that the proposed LSTM-CatBoost averaging method outperforms other benchmark models, including the individual LSTM and CatBoost prediction model. This indicates that the proposed method is more suitable for short-term weather forecasting in relevant regions of Botswana than some existing mainstream methods.

Index Terms—Internet of Things (IoT), weather forecasting, long short-term memory (LSTM), categorical boosting (CatBoost), machine learning.

I. INTRODUCTION

With the rapid development of technology, the Internet of Things (IoT) has become the most important part of our daily life. It consists of various electronic devices that connect to the internet [1], [2], such as sensors, to obtain real-time data, which can be stored and accessed by people from all over the world. In recent years, the rise and application of the IoT has brought great improvements to many aspects of human life, including weather forecasting [3].

Weather forecasting refers to the use of relevant science and technology to predict the numerical changes of relevant weather features in a location over a period of time in the future [4]. Weather forecasting plays a crucial role in people's daily lives and socioeconomic activities. Accurate weather forecasts allow us to prepare for sudden heavy rain, high temperatures, and other sudden weather changes, thereby reducing the loss of life and property caused by weather disasters [5].

Weather forecasting methods can be divided into two categories: physics-based numerical weather prediction methods

and data-driven weather forecasting methods. Physics-based numerical weather prediction methods establish mathematical and physical models based on the physical laws of the atmosphere and use supercomputers to predict future climate changes. Researchers have proposed a series of physics-based numerical methods to forecast weather for a future period [6], [7]. However, these numerical weather prediction methods have some limitations. Due to the complexity of the mathematical and physical models used, the computational load is quite large, and the solution is difficult, often requiring excessive computing resources and time [8]. Moreover, numerical weather prediction is highly dependent on the initial values and boundary conditions of the atmosphere, so accurately understanding and capturing these precise initial values and boundary conditions is crucial, as even slight deviations can lead to inaccurate weather forecasts. In addition, numerical methods are typically used for coarse-resolution large-area predictions, rather than fine-resolution predictions for specific locations [9].

On the other hand, data-driven weather forecasting methods learn patterns and relationships from historical weather data and then use the learned patterns to predict future weather variables. This includes traditional statistical methods [10] and machine learning methods [11]. Data-driven methods do not require solving complex mathematical and physical equations, which are much simpler and less costly than numerical methods, and have made significant progress in weather forecasting. Moreover, they are more suitable for predicting changes in weather indicators for specific locations. In recent years, many studies have used data-driven methods to predict relevant weather variables [12], [13], with promising results. Among the data-driven weather forecasting methods, long short-term memory (LSTM) networks is considered to have great application prospects in time series prediction because of its special structure. In recent years, many people have achieved good results in time series prediction and other tasks based on LSTM [14], [15]. However, due to its high complexity, LSTM still has the problem of overfitting, which leads to high variance error in LSTM prediction results.

In order to solve this problem, categorical boosting (CatBoost), an improved gradient boosting decision tree method, which has low bias error and is less prone to overfitting during the training process, is employed to enhance weather forecasting. By combining these two machine learning models using

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an average strategy, an LSTM-CatBoost averaging method is proposed.

Through combining the prediction results of these two machine learning models, this method can smoothen the variance error while keeping the bias error low, so that the final prediction results have both low variance error and bias error at the same time, thus having good generalization performance. The main contributions of this paper are summarized as follows:

- Aiming at the problem of weather data forecasting in tropical regions with complex and variable climates, this research combines LSTM model with other heterogeneous model based on the prediction error of the machine learning models. This kind of approach helps in achieving higher accuracy in weather forecasting, which is significant for people to prepare for sudden weather change, thereby mitigating the probable damage.
- Since conventional LSTM has high complexity and is easy to overfit, its prediction error mainly comes from the variance error. By contrast, CatBoost has low bias error and is less prone to overfitting. Motivated by this, an LSTM-CatBoost averaging method is proposed in this paper. By combining CatBoost and LSTM using an average strategy, the variance error is smoothed while maintaining a low bias error, thus getting a better generalization and forecasting performance.
- Given the IoT technology allows us to get real-time data from weather stations, we conduct experiments on the data from six IoT-based automatic weather stations located at different sites within Botswana. The experiments demonstrate that the proposed LSTM-CatBoost averaging model performs better than the other benchmark models, like the individual LSTM model and CatBoost model, indicating that it has a certain application prospect in weather forecasting.

The remainder of this paper is organized as follows. Section II describes the study area used in this research, the source and the preprocessing methods of the data. Section III introduces the methods used in this paper for weather feature prediction. Section IV presents the relevant experimental results and a detailed discussion. Finally, Section V summarizes the entire paper and provides an outlook for future work.

II. STUDY AREA AND DATA

A. Study Area

The research area of this paper is located in the Republic of Botswana. The country faces challenges such as drought, which may have an impact on the safety of its citizens' lives and property. Therefore, rapid and accurate weather forecasting is of great importance for the personal safety and property security of the people in the country.

B. Data

1) *Data Source*: Since 2017, Botswana has built a number of IoT-based automatic weather stations across the country in

order to record local weather data [16]. This weather data includes variables such as wind speed and ambient temperature. The automatic weather stations record the weather data every 15 minutes and save it in CSV format. The IoT technology makes the weather data from the weather stations accessible to people all over the world.

2) Data Preprocessing:

a) *Handling missing values*: The weather data from the Botswana automatic weather stations may have incomplete recordings due to equipment or personnel issues. There are several missing data points that need to be filled in. In this paper, we use the average of the two adjacent rows of data to fill in the missing values. If the missing values are consecutive, we handle them differently by using the previous data or the next data point to fill in the gaps.

b) *Handling outliers*: Within the continuous time period, we use an appropriately sized window to process outliers. In this paper, we adopt the 3σ principle: the probability of data distribution within $(\mu-3\sigma, \mu+3\sigma)$ is 0.9974, where μ is the average value of all the data, and σ is the standard deviation of all data. This means the probability of data falling outside this range is less than 0.003, which is very small. Therefore, we can consider data points outside this range as outliers. For these outlier values, we replace them with the average value within the window. If the length of this continuous time period is less than the window size, then we do not perform any processing. In this paper, we set the window size to 500.

c) *Normalization*: In machine learning, among the features input to the model, there are inevitably some features that are particularly large or small compared to other input features, which may increase the training time of the model and may also cause the model to fail to converge. As a result, before inputting the weather data into the adopted machine learning model, the min-max normalization is used to normalize them within a certain range, which can speed up the convergence of gradient descent to find the optimal solution, accelerate the training of the network, and can also improve the prediction accuracy of the model. The relevant formula is as follows:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

where x_{max} is the maximum value in the overall data, x_{min} is the minimum value in the overall data, x is the data that needs to be normalized, and x_{norm} is the data after x has been normalized. From the formula above, we can see that the data after normalization is within the range of 0 to 1.

d) *Constructing the dataset*: In the weather data prediction experiment carried out in this paper, the processed 2020 dataset from each IoT-based automatic weather station is divided into training set, validation set, and test set at a ratio of 6:2:2.

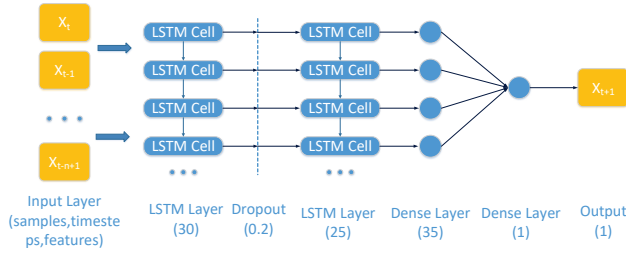


Fig. 1: The architecture of our proposed LSTM-based weather prediction model

III. PROPOSED LSTM-CATBOOST METHOD ARCHITECTURE

A. LSTM

LSTM network is a special type of Recurrent Neural Network (RNN) [17], which is an improvement based on the traditional RNN. Traditional RNNs suffer from the problems of gradient vanishing and gradient exploding [18], while LSTM solves these issues of traditional RNNs [19] and can learn long-term dependencies more effectively [20].

As shown in Fig. 1, an LSTM-based model is constructed in this paper for ambient temperature prediction. The input of the entire network is in the form of a three-dimensional tensor, represented as (Samples, Timesteps, Features). Samples refer to the size of the training set, which in our experiment is the size of the training set of each automatic weather station. Timesteps refer to the size of the time window used for predicting ambient temperature. In our method, we set it to 20, which means we use the previous 20 relevant weather indicators to predict the next one ambient temperature. Features refer to the number of weather features used in the prediction task. In this experiment, we use 7 weather variables, including ambient temperature, wind speed, ground temperature, humidity, atmospheric pressure, dew point temperature, and wet-bulb temperature, to predict the ambient temperature. Therefore, the value of Features is set to 7. Through experimentation, we set the number of units in the first LSTM layer to 30, the number of units in the second LSTM layer to 25, and the number of units in the first Dense layer to 35. We apply a dropout mechanism between the first and the second LSTM layer to help prevent overfitting during the training process. The dropout rate in this experiment is set to 0.2.

B. CatBoost

CatBoost was proposed by Prokhorenkova et al. in 2017 [21], and is an open-source machine learning library. CatBoost is an improved gradient boosting decision tree method that can be used for predictive tasks. The main problem it solves is the efficient and reasonable handling of categorical features. Additionally, CatBoost also addresses the issues of gradient bias and prediction shift, thereby reducing the occurrence of overfitting, and consequently improving the algorithm's accuracy and generalization capability [22].

C. Proposed LSTM-CatBoost Averaging Method

For machine learning models, the prediction error can be divided into three parts: bias error, variance error, and irreducible error. Since the irreducible error is hard to be reduced, the focus for machine learning models is to reduce the others. However, reducing either one will increase the other. So the bias error and variance error need to be balanced.

For the LSTM, the prediction error is mainly due to the variance error. In order to reduce the variance error of the LSTM model's prediction, it is possible combining it with other prediction technique which is less prone to overfitting.

In view of this, an LSTM-CatBoost averaging method is proposed in the following.

The general steps of the proposed LSTM-CatBoost averaging method are shown in Fig. 2.

Firstly, the weather data after relevant data preprocessing is fed into the CatBoost model and LSTM model respectively for independent training. After that, the trained CatBoost model and LSTM model are used to separately predict the next 1 data on the test set. Then, we combine the prediction results of these two methods on the test set through an averaging strategy to generate the final prediction. Next, we will use the final prediction result generated in the previous step to update the original weather sequence, which is used to predict the next 1 weather data. By continuously repeating the processes above, the proposed LSTM-CatBoost averaging method can realize the prediction of the next n data.

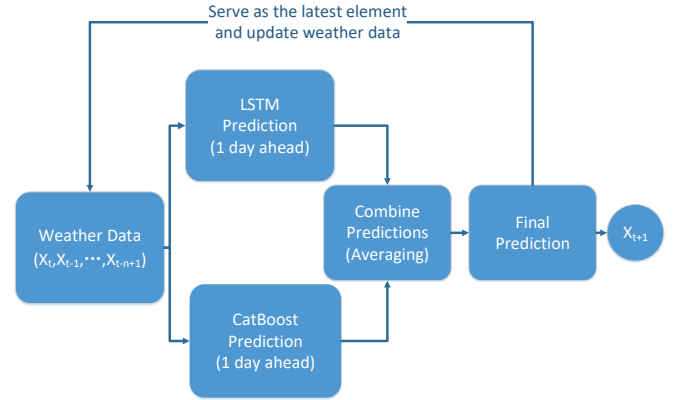


Fig. 2: Proposed LSTM-CatBoost averaging method

Remark 1: The proposed LSTM-CatBoost averaging method applies an averaging strategy to combine the two heterogeneous models, namely the CatBoost model and the LSTM model. By utilizing the averaging strategy, the variance error of the LSTM model's prediction can be reduced while maintaining low bias error. As a result, the proposed LSTM-CatBoost averaging method has both low variance error and low bias error in principle, thereby having a good generalization and fitting performance at the same time, which further improves the prediction results.

IV. SIMULATION RESULTS

A. Data Selection

In this experiment, we selected 6 IoT-based automatic weather stations with relatively good original data records and fewer missing values, numbered 68024, 68030, 68038, 68148, 68226, and 68328.

B. Evaluation Metrics

In this experiment, we assume the true value is $y = y_1, y_2, \dots, y_n$ and the predicted value is $\hat{y} = \hat{y}_1, \hat{y}_2, \dots, \hat{y}_n$. In terms of evaluation metrics, we have chosen mean absolute percentage error (MAPE) and mean absolute error (MAE) as the commonly used evaluation metrics to assess the superiority of the method we proposed.

1) *MAPE*: Mean absolute percentage error [23]. The range of this value is $[0, +\infty)$. The larger the error, the larger this value. The calculation formula is as follows:

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (2)$$

2) *MAE*: Mean absolute error [24]. The range of this value is $[0, +\infty)$. The larger the error, the larger this value. The calculation formula is as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (3)$$

C. Experimental environment

The experimental environment of this paper is Windows 10 operating system, python 3.8.8. The LSTM deep neural network and convolutional neural network (CNN) are implemented using keras 2.12.0, the support vector machine (SVM) is implemented using Scikit-learn 1.3.2, CatBoost is implemented using catboost 1.2.2, light gradient boosting machine (LightGBM) is implemented using lightgbm 4.3.0.

D. Result Analysis

The experiment is divided into single-step prediction and multi-step prediction. Single-step prediction is to use the seven weather features of the first 20 data in the weather data, including ambient temperature, wind speed, ground temperature, humidity, atmospheric pressure, dew point temperature and wet bulb temperature, to predict the ambient temperature in the next data. The multi-step prediction uses the same data as the single-step prediction, which predicts the next five pieces of data of the ambient temperature.

1) *Single-Step Prediction*: The following six pictures in Fig. 3 respectively display the prediction results on the test set of the proposed method in six selected IoT-based automatic weather stations. The actual ambient temperature is plotted using a green line, the ambient temperature predicted by the proposed method is plotted using a red line.

From the six images shown in Fig. 3 below, we can see that apart from a slight underperformance in the peak regions,

the proposed method can effectively complete the task of predicting the next 1 data.

It should be noted that the slight underperformance in the peak regions occurs in all the methods in this experiment. This may be because extreme high temperatures are rare, and in order to reduce the loss during training, the model is inclined to give temperatures that are not so high at the peak [5]. It may be useful to combine some physics-based numerical weather prediction methods [6], [7], through which more accurate predictions can be made in the peak regions by analyzing the relevant meteorological conditions.

The following Table I to Table VI compare the evaluation metrics of the proposed method and other benchmark methods on the test set for predicting the future data at the six selected IoT-based automatic weather stations.

TABLE I: The comparison of relevant indicators for various methods at the 68024 station.

	LSTM-CatBoost	LSTM	CatBoost	SVM	LightGBM	CNN
MAE	0.440	0.565	0.514	2.026	0.767	0.520
MAPE	0.01674	0.02247	0.01820	0.07199	0.02605	0.01930

TABLE II: The comparison of relevant indicators for various methods at the 68030 station.

	LSTM-CatBoost	LSTM	CatBoost	SVM	LightGBM	CNN
MAE	0.438	0.455	0.608	1.902	0.740	0.610
MAPE	0.01593	0.01834	0.02061	0.06695	0.02455	0.02177

TABLE III: The comparison of relevant indicators for various methods at the 68038 station.

	LSTM-CatBoost	LSTM	CatBoost	SVM	LightGBM	CNN
MAE	0.711	0.765	0.744	3.516	0.828	0.891
MAPE	0.02963	0.03336	0.02965	0.12749	0.03136	0.03759

TABLE IV: The comparison of relevant indicators for various methods at the 68148 station.

	LSTM-CatBoost	LSTM	CatBoost	SVM	LightGBM	CNN
MAE	0.279	0.292	0.321	1.595	0.438	0.333
MAPE	0.01148	0.01227	0.01301	0.06463	0.01678	0.01395

TABLE V: The comparison of relevant indicators for various methods at the 68226 station.

	LSTM-CatBoost	LSTM	CatBoost	SVM	LightGBM	CNN
MAE	0.410	0.480	0.412	3.890	0.715	0.659
MAPE	0.01550	0.01814	0.01576	0.14802	0.02508	0.02673

TABLE VI: The comparison of relevant indicators for various methods at the 68328 station.

	LSTM-CatBoost	LSTM	CatBoost	SVM	LightGBM	CNN
MAE	0.510	0.553	0.534	3.471	0.687	0.608
MAPE	0.02654	0.02981	0.02662	0.13936	0.02991	0.03065

From Table I to Table VI above, we can see that although the prediction error of diverse methods may differ in different IoT-based automatic weather stations due to the different climate, the proposed LSTM-CatBoost averaging method still outperforms other benchmark methods in the prediction of all automatic weather stations.

The following Table VII and Table VIII show the percentage gain of the proposed method compared to the benchmark

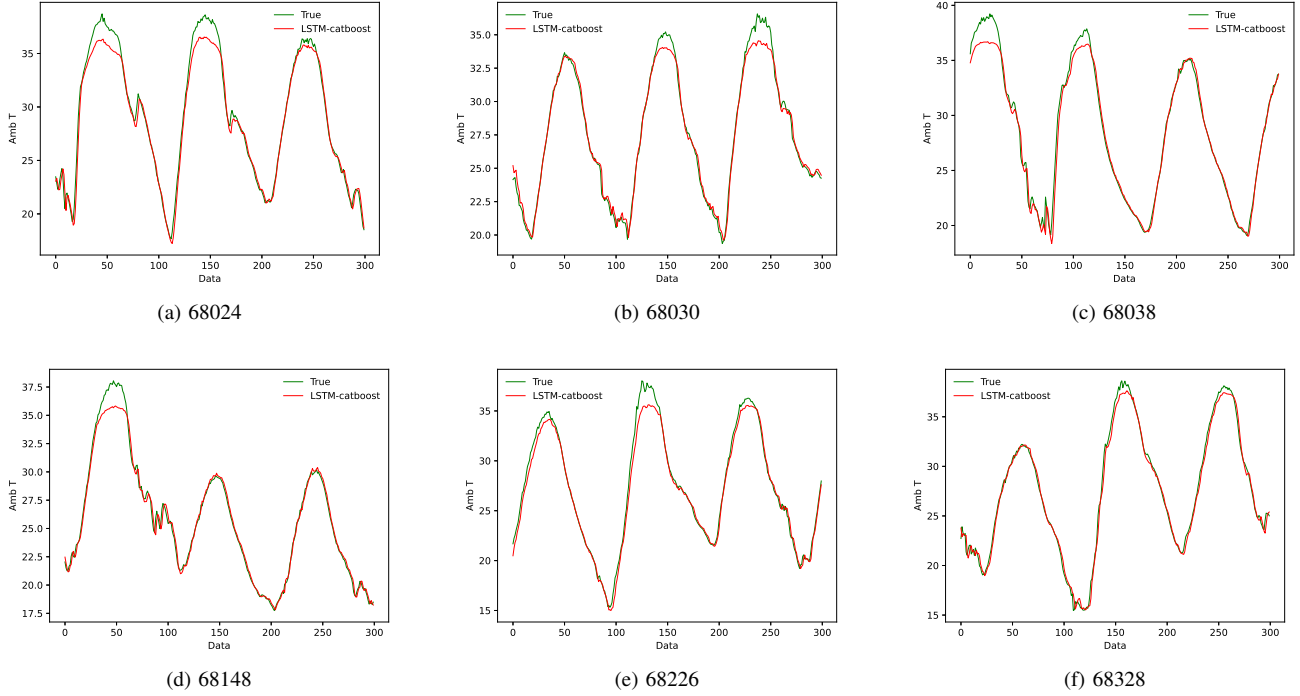


Fig. 3: The Single-Step prediction results of the proposed method at each automatic weather station

TABLE VII: Percentage improvements of MAE of the proposed method relative to other methods at each station.

	LSTM	CatBoost	SVM	LightGBM	CNN
68024	22.12	14.40	78.28	42.63	15.38
68030	3.74	27.96	76.97	40.81	28.20
68038	7.06	4.44	79.78	14.13	20.20
68148	4.45	13.08	82.51	36.30	16.22
68226	14.58	0.49	89.46	42.66	37.78
68328	7.78	4.49	85.31	25.76	16.12

TABLE VIII: Percentage improvements of MAPE of the proposed method relative to other methods at each station.

	LSTM	CatBoost	SVM	LightGBM	CNN
68024	25.50	8.02	76.75	35.74	13.26
68030	13.14	22.71	76.21	35.11	26.83
68038	11.18	0.07	76.76	5.52	21.18
68148	6.44	11.76	82.24	31.59	17.71
68226	14.55	1.65	89.53	38.20	42.01
68328	10.97	0.30	80.96	11.27	13.41

methods in the relevant evaluation metrics, which strengthen the above conclusion using precise calculations.

It is shown that the average prediction results of the proposed LSTM-CatBoost averaging method has a certain degree of accuracy improvement compared to the prediction results of other benchmark methods, which means that the proposed method is more suitable for weather forecasting in relevant areas of Botswana than other benchmark methods.

The above results indicate that the proposed method is able to achieve more accurate predictions of ambient temperature at various IoT-based automatic weather stations in Botswana,

with good generalization performance in single-step prediction.

2) *Multi-Step Prediction*: Take the experimental results of the IoT-based automatic weather station numbered 68328 as an example. The following Fig. 4 and Fig. 5 separately compare the MAPE and MAE of the multi-step prediction of the proposed LSTM-CatBoost averaging method and some benchmark methods in this IoT-based automatic weather station.

We can find that the prediction error of all methods increases with the increase of the number of predicted data, which is not difficult to understand. On the one hand, the predictability of ambient temperature decreases with the increase of the number of predicted data. On the other hand, since the predicted data is used to update the original weather sequence, the prediction error propagates and accumulates with the increase of the number of predicted data.

From Fig. 4 and Fig. 5 below, we can see that the prediction error of the proposed LSTM-CatBoost averaging method in multi-step prediction is smaller than that of LSTM and CatBoost alone. This shows that the proposed method is superior to the other methods in multi-step prediction.

V. CONCLUSIONS

The development and application of IoT has profoundly affected many aspects of human life, such as weather forecasting. Accurate weather forecasting can help people reduce the loss caused by weather changes. To achieve accurate weather forecasting in Botswana, a method that combined the CatBoost and the LSTM by using an average strategy, namely the LSTM-CatBoost averaging method, was proposed. And experiment was carried out using the data from the

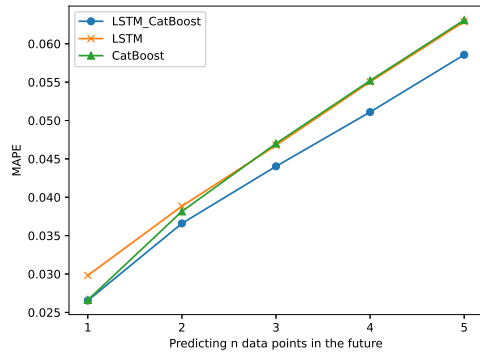


Fig. 4: MAPE comparison of multi-step predictions using the LSTM-CatBoost, LSTM, CatBoost at the 68328 station

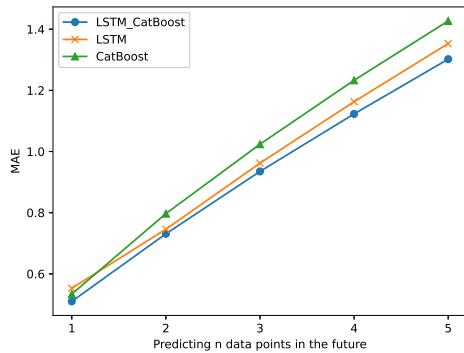


Fig. 5: MAE comparison of multi-step predictions using the LSTM-CatBoost, LSTM, CatBoost at the 68328 station

IoT-based automatic weather stations within Botswana. The results demonstrated that this method outperformed the other benchmark methods, like the standalone LSTM and CatBoost, in predicting the future ambient temperature. As for joint training of multiple automatic weather stations, such as federated learning, will be delegated to our future work.

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